Research Paper Assistant using Retrieval-Augmented Generation (RAG)

Multimodal, Multi-Paper Scientific QA with Grounded Visual and Claim Verification

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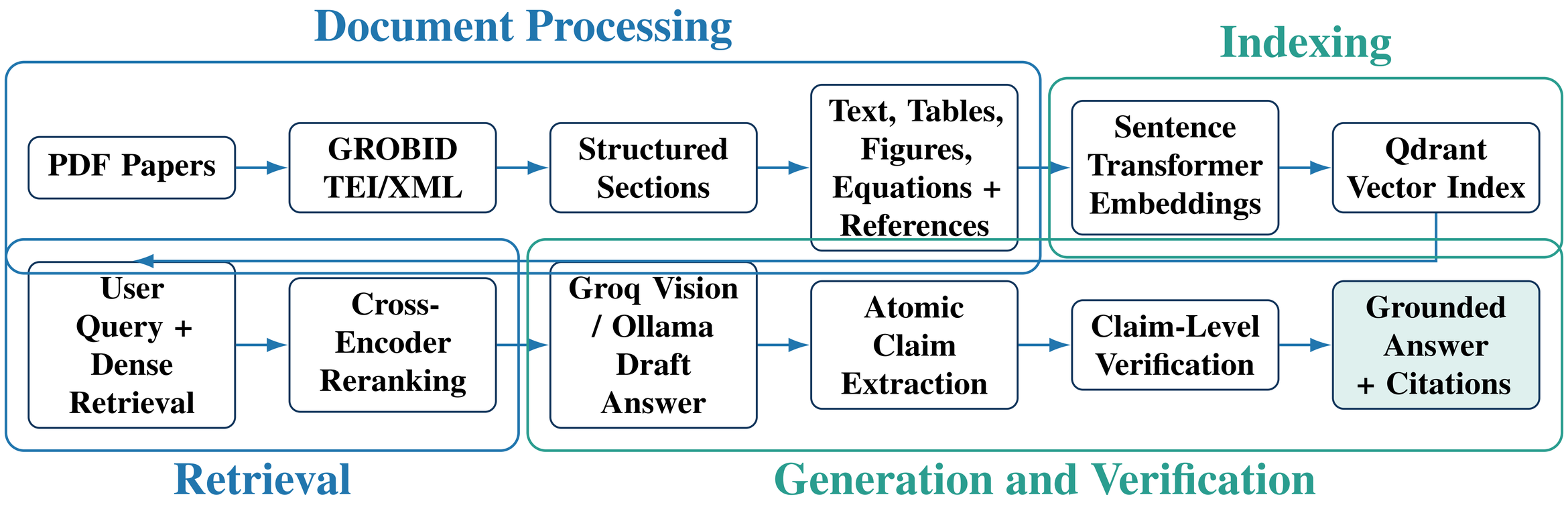


Figure 1: End-to-end architecture of the multimodal, evidence-first RAG pipeline.

Abstract and objective

Researchers must locate evidence across long, structured papers. General-purpose language models can answer from outside knowledge or introduce unsupported claims.

Objective: answer single-paper and cross-paper questions using only retrieved evidence, with citations and claim-level filtering before synthesis.

- Ingest multiple scientific PDFs.
- Preserve text, tables, figures, equations, references, and provenance.
- Retrieve and rerank relevant passages.
- Compare evidence across named papers and requested aspects.
- Answer questions from both text and cropped paper figures.
- Remove claims not supported by retrieved excerpts.

Scientific PDF ingestion

GROBID converts PDFs into structured TEI/XML. The parser retains title, authors, abstract, sections, paragraphs, tables, formulas, bibliography entries, figure captions, and PDF crop coordinates.



PyMuPDF uses GROBID coordinates to crop retrieved figures from the original PDF. The same image is displayed with the answer and sent to Groq's vision-capable model for grounded visual interpretation.

Section-aware semantic indexing

Each chunk retains:

- paper title and document name,
- section name and chunk ID,
- evidence type, label, and caption,
- figure crop coordinates when available.

Embedding model	all-MiniLM-L6-v2
Vector dimension	384
Vector database	Qdrant
Similarity	Cosine similarity

Dense vectors support semantic search while metadata preserves where every passage originated.

Retrieval and reranking



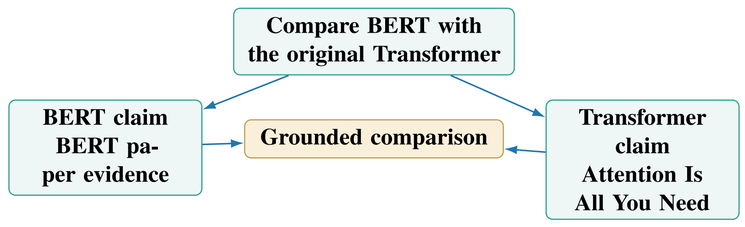
Dense retrieval supplies candidates. The `ms-marco-MiniLM-L-6-v2` cross-encoder then scores each question-passage pair and improves evidence ordering.

Technology stack

Language	Python
Parsing	GROBID, TEI/XML, PyMuPDF
Embeddings	Sentence Transformers
Vector store	Qdrant
Reranking	Cross-Encoder
Generation	Groq Qwen3.8 27B Vision; Ollama Llama 3.1 8B
Interface	Streamlit
Workflow	Docker, Jupyter

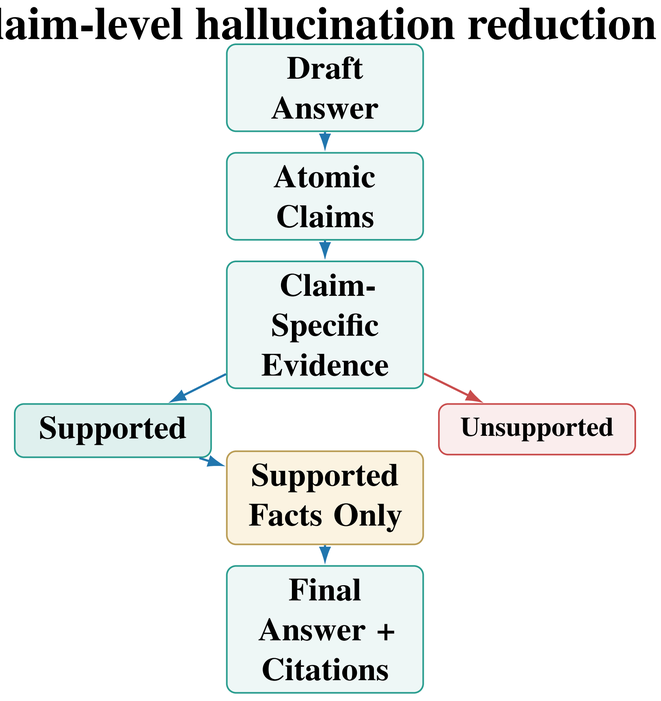
Multi-paper comparison

Naive global search can over-represent one paper. Broad comparison questions are decomposed into objectives, methods, datasets, results, limitations, and contributions. Each aspect is retrieved independently for every selected paper.



Routing signals: paper filters, aspect decomposition, balanced evidence, atomic claims, entity detection, primary-subject identification, and claim-specific retrieval. Dataset and result searches retrieve extra passages to recover named benchmarks and numerical metrics.

Claim-level hallucination reduction



The final synthesizer receives verified facts rather than the draft. Source IDs, paper scope, entity compatibility, duplicates, and malformed partial verdicts are validated before a fact can reach the answer.

Verification example

Rejected draft claim

"The original Transformer uses constrained self-attention."

The evidence describes masking in the decoder; it does not establish that the entire architecture uses constrained self-attention. **Removed.**

Supported claim

"BERT uses bidirectional self-attention."

Explicitly supported by the BERT model-architecture evidence. **Retained.**

Grounded comparison example

Question: Compare BERT with the original Transformer architecture.

BERT

- Multi-layer bidirectional Transformer encoder.
- Uses bidirectional self-attention.
- Unified architecture across tasks.

Original Transformer

- Attention-based sequence-transduction model.
- Encoder-decoder architecture.

Key difference: BERT is encoder-based; the original Transformer has both encoder and decoder components.
[Source: BERT, Model Architecture] [Source: Transformer, Model Architecture]

Evaluation framework implemented

The repository includes deterministic unit tests and a reusable end-to-end evaluation framework. No completed benchmark run is stored in `evaluation/results/`, so the poster does not invent performance scores.

Retrieval evaluation

- Hit@1, Hit@3, Hit@5, and mean reciprocal rank.
- Vector retrieval before vs. after reranking.
- Paper and section matching.
- Comparison-paper coverage.

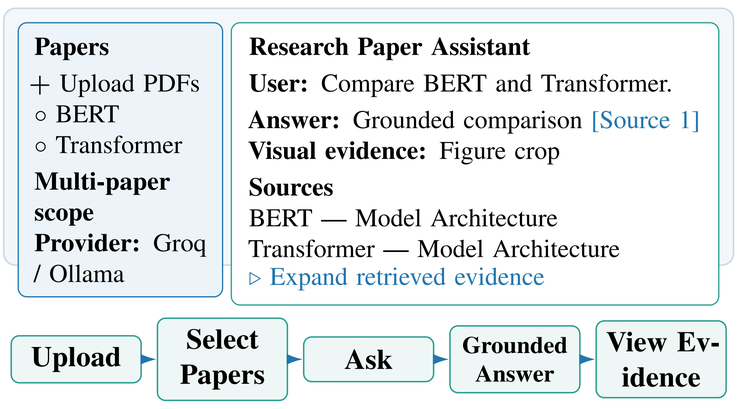
Grounding evaluation

- Fully, partially, and unsupported claim rates.
- Citation validity and rejected-claim leakage.
- Expected-fact coverage and abstention accuracy.

18 Evaluation questions	**50** Passing unit tests
4 Structured evidence types	**0** Fabricated metrics

Streamlit User Interface

The Streamlit interface operates the complete pipeline: upload and replace indexed papers, choose a single- or multi-paper scope, select local Ollama or Groq API generation, ask questions, and inspect textual and visual evidence.



Transparent answers: citations expose paper, section, evidence type, label, and excerpt. Retrieved figures appear directly below the answer and again inside Sources. The UI reports single-paper, multi-paper, or library-wide scope.

Implemented contributions

- Structure-aware scientific PDF ingestion.
- Typed indexing of paragraphs, tables, equations, figures, and references.
- Coordinate-based figure cropping and multimodal Groq inference.
- Dense retrieval plus cross-encoder reranking.
- Aspect-balanced, paper-aware multi-paper evidence routing.
- Atomic claim extraction and source-scope verification.
- Citation-grounded synthesis, safe re-index replacement, and evaluation.

Perspectives

- Groq sends selected excerpts and figure crops to a cloud service.
- Figures without GROBID coordinates remain caption-only evidence.
- Dense retrieval can miss highly specific technical evidence.
- Verified mode increases API calls and inference time.
- Next: hybrid retrieval, visual-table OCR, and a larger benchmark run.

Conclusion

The system combines typed scientific-document processing, balanced multi-paper retrieval, visual figure understanding, cloud or local generation, citations, and claim-level verification to answer from evidence researchers can inspect.

References

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[2] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.
[3] J. Devlin et al. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.